🌌 ZenWall | Signature Atmospheric Noir Wallpapers

ZenWall is a premium, high-performance SaaS platform that leverages advanced AI to create minimalist, atmospheric noir wallpapers. Designed for focus and digital serenity, ZenWall transforms simple prompts into cinematic, high-definition masterpieces.

🚀 Key Features

- **ZenAI Flux Engine**: Custom-tuned prompt engineering for "Signature Atmospheric Noir" aesthetics—wide-angle, high-contrast, and deep black backgrounds.
- **Immersive Digital Vault**: A private user dashboard to sync, curate, and revisit your entire high-res art collection across devices.
- **Cinematic Detail View**: High-definition viewer with ambient background glows and detailed prompt transparency.
- **Stripe Payments**: Secure, one-time credit top-ups with full Stripe Checkout integration and webhook verification.
- **Premium Security**:
  - Strict **Zod** validation on both client and edge functions.
  - Multi-tier **Rate Limiting** to prevent API abuse and protect credits.
  - **Atomic SQL Credits**: Thread-safe credit deduction directly in the database layer.
- **Code Quality**: Automated testing with **Jest**, code formatting with **Prettier**, and strict linting with **ESLint**.

📸 Feature Showcases

1. Interactive Landing Page

> Atmospheric noir aesthetic with 3D scroll effects. Landing Page Showcase

2. AI Generation Interface

> Simple, minimalist prompting with Zod-validated inputs. Generation Interface

3. Private Collection Vault

> Manage your generated history with ease. Dashboard Vault

4. Secure Stripe Payments

> Global payment processing for credit top-ups. Stripe Checkout

5. Mobile-First Experience

> Fully responsive noir interface optimized for every screen size.

Mobile Dashboard	Mobile Viewer
Mobile Screen 1	Mobile Screen 2

🛠️ Tech Stack

- **Framework**: Next.js 16+ (App Router & React Compiler)
- **Database & Auth**: Supabase
- **State Management**: Zustand (Persistent Auth & Hydration)
- **Payments**: Stripe API
- **AI Engine**: Hugging Face (FLUX.1-schnell)
- **Image Pipeline**: Cloudinary (Signed Uploads)
- **Animations**: Framer Motion
- **Styling**: Tailwind CSS 4.0
- **Notifications**: Sonner (High-performance stacked toasts)
- **Validation**: Zod
- **Testing**: Jest & React Testing Library
- **Quality Control**: ESLint & Prettier

📦 Getting Started

1. Clone & Install

```
git clone https://github.com/yourusername/zenwall.git
cd zenwall
npm install
```

2. Environment Variables

Create a `.env.local` file:

```
NEXT_PUBLIC_SUPABASE_URL=your_url
NEXT_PUBLIC_SUPABASE_ANON_KEY=your_key
SUPABASE_SERVICE_ROLE_KEY=your_service_key
HUGGINGFACE_TOKEN=your_hf_token
CLOUDINARY_CLOUD_NAME=your_cloud_name
CLOUDINARY_API_KEY=your_api_key
CLOUDINARY_API_SECRET=your_api_secret
STRIPE_SECRET_KEY=your_stripe_secret
STRIPE_WEBHOOK_SECRET=your_webhook_secret
```

3. Run Quality Checks

```
npm run lint    # Check for code issues
npm run format  # Format code with Prettier
npm test        # Run Jest test suites
```

4. Start Development

```
npm run dev
```

🛡️ Production Readiness Checklist

- ✅ Zod Input Sanitization (Client & Server)
- ✅ Anti-Spam Rate Limiting (10s Cooldown)
- ✅ Stripe Webhook Security Verification
- ✅ SEO & Meta Tag Hardening
- ✅ Terms & Privacy Compliance
- ✅ Automated Jest Test Coverage

📄 License

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